

TRƯỜNG MINH HOÀNG

AI Engineer Intern | Computer Vision

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Summary

Aspiring AI Engineer Intern seeking to deepen expertise in Deep Learning, especially in Computer Vision areas such as image segmentation, face recognition, behavior analysis, and face anti-spoofing. Eager to contribute to real-world AI projects, learn from experienced engineers, and grow technical capabilities in a professional environment.

Education

Hanoi University of Science and Technology

Mathematics and Computer Science

CPA: 2.82/4; GPA: 3.85/4. Fourth-year student majoring in Mathematics and Computer Science.

Academic Encouragement Scholarship, Semester 2025.2

2022 – Present

Hanoi, Vietnam

Skills

Languages / Tools: Python, Java, SQL, Git, Docker

Backend: Node.js, Express.js, Spring Boot

Frontend: React, TypeScript, Vite

Databases: PostgreSQL, Redis.

AI/ML: PyTorch, Scikit-learn, YOLOv8, OpenCV, Computer Vision, RAG, LangChain

Work Experience

VNPT Net Corporation

Intern

2025

Hanoi, Vietnam

- Contributed to the development and optimization of an internal network infrastructure management system.
- Worked with Java, Spring Boot, and Oracle Database in an enterprise environment.
- Participated in maintenance, testing, and performance improvements of system features.

VNPT AI Company

AI Engineer Intern

2026

Hanoi, Vietnam

- Studied core concepts in Machine Learning, Deep Learning, and Computer Vision.
- Participated in building image recognition models and data analysis workflows.
- Implemented computer vision projects using CNNs, YOLOv8, and OpenCV.
- Explored and applied AI model optimization techniques for real-world applications.
- Contributed to AI systems integrated with large language models (LLMs) and RAG.

Selected Projects

Football Player Re-Identification System

Personal Project

- Technologies: Python, YOLOv8, BoT-SORT, PyTorch, OpenCV.
- Built a Detection → Tracking → ReID pipeline to track players and maintain consistent identities throughout football match videos.
- Used YOLOv8 for player detection in each frame and integrated BoT-SORT for multi-object tracking across consecutive frames.
- Developed a ReID model to extract player appearance features, reducing ID switches caused by occlusion or players leaving the frame.
- Performed data preprocessing, model training, evaluation, and real-time player ID visualization on output videos.
- Result: improved identity persistence when players were occluded or reappeared, reducing ID switches during tracking.

Food Ingredient Recognition from Images

Personal Project

- Technologies: Python, YOLOv8, PyTorch, OpenCV.
- Built a food ingredient recognition system from images to support automatic recipe recommendation and integration into

CookSmart.

- Collected, cleaned, and labeled data; trained YOLOv8, tuned hyperparameters, and applied data augmentation to improve generalization.
- Built a real-time inference pipeline with OpenCV and connected recognition results to recipe search and recommendation features.
- Result: recognized 65 ingredient categories, achieved over 80% mAP@50 on the test set, and kept inference time under 100 ms per image.

RAG-Based Cooking Assistant Chatbot

Personal Project

- Technologies: Python, LangChain, RAG, Vector Database, LLM, FastAPI.
- Built an AI chatbot for recipe lookup, ingredient consultation, and cooking guidance for the CookSmart system.
- Designed a RAG pipeline including intent detection, retrieval, and response generation; built a knowledge base from recipes, ingredients, and cooking instructions.
- Generated embeddings, implemented semantic retrieval with a vector database, and developed APIs to integrate the chatbot into the system.
- Result: improved answer accuracy, reduced hallucinations, and effectively supported questions about dishes, ingredient substitutions, and cooking processes.

CookSmart – Recipe Recommendation Website

Personal Project

- Built a full-stack web application that recommends recipes based on ingredients entered by users.
- Backend: Node.js, Express.js, PostgreSQL, Sequelize, JWT, bcrypt, Redis, Elasticsearch, Socket.IO.
- Frontend: React, TypeScript, Vite, TailwindCSS, React Query, Context API, Axios, React Router.
- Added rate limiting, Winston logging, and API documentation with Swagger/OpenAPI 3.0.

Certifications & Activities

Certifications: Neural Networks & Deep Learning (Coursera – DeepLearning.AI), TOEIC 580

Activities: Vice President, HUST Debate Club; Head of Organizing Committee, HUST Debate Open Champion; Deputy Head of Organizing Committee, talk show “Giao Lo”.